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Meteorology shapes urban particle nucleation, while pollution-related factors modulate growth



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Urban new particle formation (NPF) is shaped by meteorology and pollution, yet their stage-specific roles across particle sizes remain difficult to quantify. Here we combine size-resolved particle number size distributions from polluted Beijing and cleaner coastal Qingdao with XGBoost modeling and SHAP attribution to resolve environmental contributions across 10–500 nm particles and three modes. The models reproduced observed particle number concentrations with high skill (mean test $R^2 = 0.95$ in Beijing and 0.88 in Qingdao), enabling size- and mode-specific attribution of model predictions. Meteorological variables, especially relative humidity and solar radiation, accounted for the largest share of nucleation-mode variability (~56–58%), and their contributions intensified during NPF events, reaching ~70% in Qingdao. By contrast, pollution-related variables gained importance in growth-related size modes, with city-specific differences. These results show that meteorological conditions were most strongly associated with observable nucleation-mode bursts, whereas pollution-related conditions gained importance during subsequent growth, providing observational benchmarks for urban aerosol formation and size-resolved aerosol-model sensitivities.

New particle formation (NPF) converts gaseous precursors into atmospheric particles and is an important source of urban ultrafine particles and climate-relevant aerosol populations^{1–5}. Although NPF is episodic, its impact on urban particle burdens can be substantial: long-term urban observations have reported localized formation-and-growth processes on about 7% of days, with ultrafine particle number concentrations reaching $60 \times 10^3 \text{ cm}^{-3}$ ⁶. The atmospheric relevance of NPF therefore depends not only on the appearance of nanometer-sized particles, but also on their persistence and growth into larger size ranges. NPF is thus a coupled formation-and-growth process, not only an event-occurrence problem.

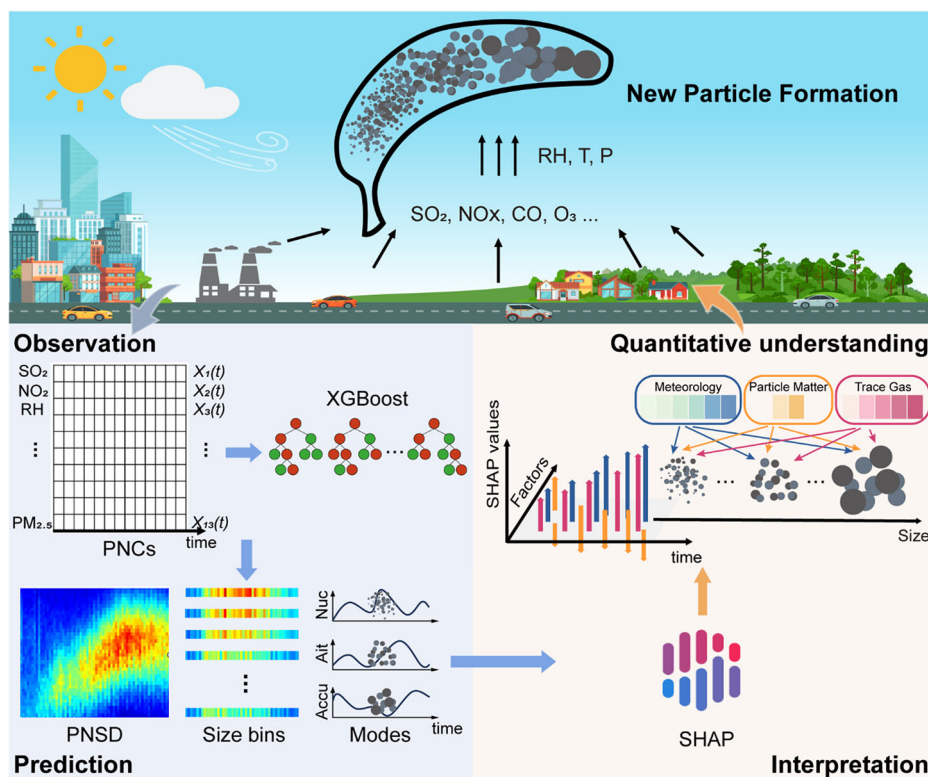
The formation and growth stages are influenced by related but distinct processes. Nucleation involves molecular clusters formed from precursors such as sulfuric acid, amines and low-volatility organic compounds, whereas subsequent growth depends on condensable vapors, coagulation and interactions with pre-existing particles^{7–13}. This distinction is central because newly formed particles must grow rapidly enough to avoid scavenging by larger aerosols, especially in the smallest size

ranges, before they can contribute to larger aerosol populations^{12,13}. Field observations have linked NPF events to strong solar radiation, low relative humidity, low background aerosol loading and changes in trace gases such as NO_2 , O_3 , and SO_2 ^{14–18}. These observations show that both meteorology and atmospheric composition are relevant, but they do not directly quantify how their relative roles vary across particle sizes and stages. Large observational analyses have identified several common environmental conditions associated with NPF. For example, a study covering more than 85 site-years at 16 European sites identified 1952 NPF events and examined their frequency, formation rate and growth rate in relation to meteorology and atmospheric composition¹⁴. Solar radiation, temperature and pressure were generally positively associated with event frequency, whereas relative humidity was generally negatively associated; however, exceptions occurred among sites, and both the magnitude and direction of some relationships varied with local conditions. These findings show that the field has moved beyond asking whether meteorology and composition matter. The more specific question is how their

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Fig. 1 | Observation-to-attribution workflow for size-resolved urban particle formation. Time-resolved meteorological variables, trace gases, and background particulate matter were used as predictors in XGBoost models trained for individual particle-size bins and operational particle modes. Model outputs were interpreted with SHapley Additive exPlanations (SHAP) to attribute predicted particle number concentrations (PNCs) to individual predictors and grouped predictor classes. The workflow resolves how meteorological and pollution-related contributions vary with particle size, particle mode and event condition.



contributions are partitioned across particle sizes, particle modes and urban environments with different pollution backgrounds.

This question is difficult to answer because urban NPF is nonlinear, size-dependent and embedded in covarying atmospheric conditions. Meteorological variables can affect photochemistry, humidity-dependent sinks and dilution, while pollution-related variables can serve as proxies for precursor availability, oxidant chemistry, emissions and pre-existing particle sinks^{19–26}. Event classification, correlation analysis and formation- or growth-rate diagnostics provide essential constraints on NPF occurrence and development^{27,28}. However, these approaches usually summarize NPF using event labels or a small number of process metrics, and are less suited to assigning measured environmental contributions continuously across the 10–500 nm size range and across nucleation, Aitken and accumulation modes.

Interpretable machine learning provides one route to this attribution problem, provided that its outputs are interpreted within the limits of observational data. Nonlinear models can reproduce complex relationships among atmospheric variables, while interpretation methods can decompose model predictions into variable-level contributions^{29–32}. Recent studies using physically informed or interpretable learning have shown how predictive models can be linked to process-relevant variables rather than used as black boxes^{33–36}. Here, SHapley Additive exPlanations (SHAP) is used not as a causal test, but as a quantitative tool for comparing how measured meteorological and pollution-related variables contribute to modeled particle number concentrations across sizes and atmospheric states^{31,32}.

Here we apply a size- and stage-resolved explainable machine-learning approach to urban NPF using particle number size distributions measured in two contrasting urban environments in China: polluted Beijing and cleaner coastal Qingdao (Fig. S1). XGBoost models were trained to predict particle number concentrations across 10–500 nm and within nucleation, Aitken and accumulation modes³⁷; SHAP was then used to attribute model predictions to meteorological variables, trace gases and background particulate matter. This approach allows us to ask how environmental contributions shift from nucleation to growth, how they differ between urban backgrounds and how they change during NPF event periods. We find that meteorological variables, especially relative humidity and solar radiation, account for the largest share

of model-explained nucleation-mode variability, whereas pollution-related variables gain importance in Aitken and accumulation-mode particles. By resolving these contributions across particle size, formation stage and event condition, this study provides an interpretable basis for urban aerosol formation and observational benchmarks for size-resolved aerosol-model sensitivities.

Results

Size-resolved models provide a reliable basis for interpretable attribution

Urban NPF rapidly redistributes particle number concentrations (PNCs) across a broad size range, making size-resolved prediction a necessary first step before environmental attribution. We therefore trained separate models for PNCs across 10–500 nm using minute-scale observations of 13 environmental variables, including meteorological parameters, trace gases and background particulate matter (Fig. 1 and Table S1). Four modeling approaches were evaluated using the same predictor set: Ridge regression, fully connected neural networks, Random Forest and XGBoost. XGBoost provided the best balance of predictive skill and stability, with mean test R^2 values averaged across size-bin models of 0.95 in Beijing and 0.88 in Qingdao, and lower RMSE values across most particle sizes than the other models (Tables S2 and S3).

Prediction errors were largest in size ranges associated with rapid nucleation and early growth, particularly 9–60 nm in Beijing and 20–80 nm in Qingdao (Figs. S2 and S3). This pattern is expected because these size bins exhibit the strongest short-term variability during NPF events. Even under these conditions, the XGBoost models reproduced the observed particle number size distributions with high fidelity (Figs. S4 and S5). On active NPF days, absolute errors in individual size bins remained below 1000 cm⁻³ in Beijing and 5000 cm⁻³ in Qingdao. Given that peak event-day PNCs were typically $\sim 5 \times 10^4$ – 8×10^4 cm⁻³, these errors correspond to approximate relative errors below $\sim 2\%$ and $\sim 8\%$, respectively. These results show that the models capture the main size-resolved PNC variability during both background and event conditions, providing a suitable basis for SHAP-based attribution of model predictions.

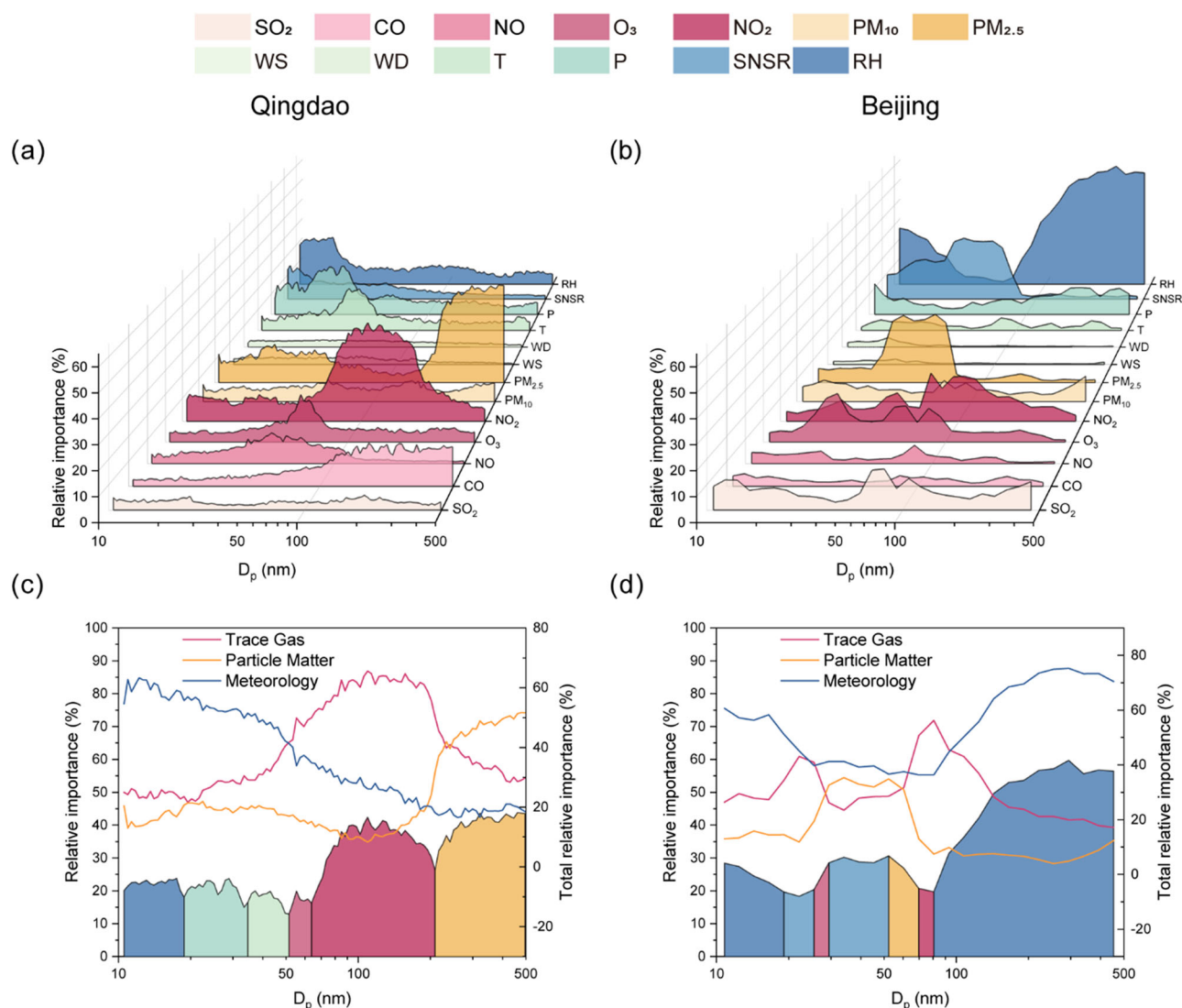


Fig. 2 | Size-resolved environmental attribution of particle number concentrations in Qingdao and Beijing. **a, b** Show SHAP-based relative importance of individual predictors for particle number concentration (PNC) across particle-size bins in Qingdao and Beijing, respectively. **c, d** Show the dominant predictor in each

size bin and the cumulative relative importance of three predictor groups: meteorological variables, trace gases and background particulate matter. Relative importance was calculated from mean absolute SHAP values and is expressed as a percentage of the total attribution within each size-bin model.

Meteorological and pollution-related contributions partition strongly by particle size

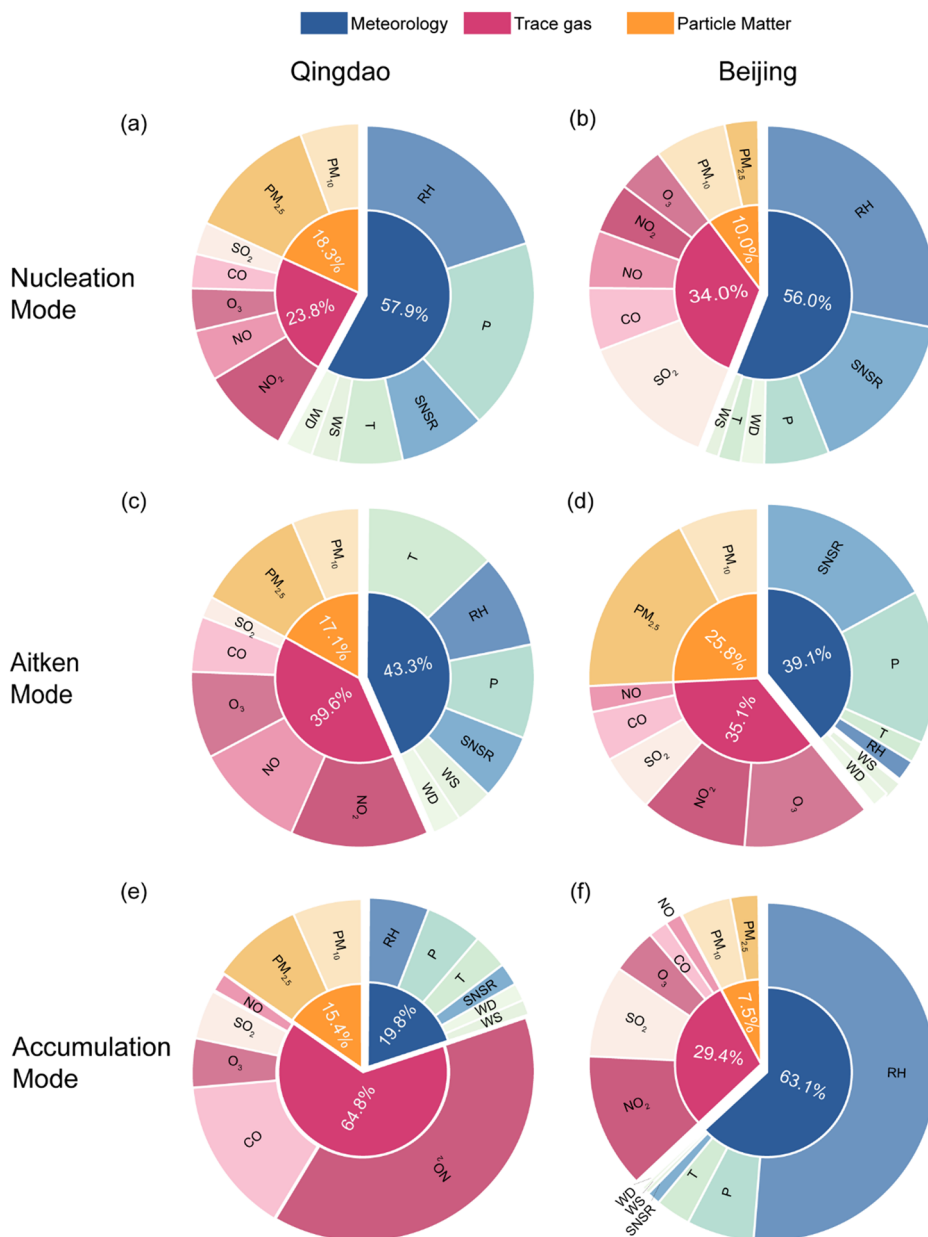
We next applied SHAP to each size-resolved XGBoost model to quantify how the 13 environmental variables contributed to modeled PNC variability across 10–500 nm (Fig. 2). The attribution patterns showed strong size dependence and clear differences between the two cities. In Qingdao, relative humidity (RH) was a leading contributor to the smallest particles, accounting for ~20–24% of model-explained variability in the 10–25 nm range, but its contribution dropped to <8% above 25 nm. In contrast, NO₂ showed a pronounced contribution between 60 and 180 nm, reaching ~40%, while remaining <10% outside this interval. In Beijing, RH also contributed strongly at small sizes, with relative importance of ~10–27% below 20 nm, but it remained influential over a wider size range and reached 56.8% across 100–500 nm. SO₂ was negligible in Qingdao, with contributions ≤5% across sizes, but exceeded ~10% near 15 nm and again around 80 nm in Beijing.

To clarify how individual variables combine into broader environmental categories, we grouped predictors into meteorological variables, trace gases and background particulate matter. The grouped SHAP contributions revealed a consistent size-dependent partitioning.

Meteorological variables accounted for the largest contribution at the smallest sizes, reaching ~55–80% below 50 nm. Pollution-related variables, including trace gases and particulate matter, became more important in larger size ranges, especially near the transition from Aitken to accumulation-mode sizes. In Qingdao, RH, pressure and temperature were the leading contributors below 50 nm, whereas O₃, NO₂ and PM_{2.5} became more important above 50 nm. In Beijing, RH and solar radiation remained important across a broader size range, while O₃, NO₂ and PM_{2.5} contributed most strongly within a narrower mid-size window. These results show that the relative importance of meteorology and pollution-related variables is not fixed, but shifts systematically with particle size.

Together, the size-resolved results show that urban PNC variability cannot be explained by a single dominant environmental factor. Instead, the attribution pattern changes with particle diameter: meteorological variables are most influential in the smallest size range associated with nucleation and early growth, whereas pollution-related variables gain importance as particles grow into larger modes. This size dependence provides the first layer of evidence that meteorological and pollution-related influences partition differently across the NPF sequence.

Fig. 3 | Mode-resolved environmental attribution of particle number concentrations in Qingdao and Beijing. **a, c, e** Show SHAP-based relative importance of individual predictors for nucleation-mode, Aitken-mode and accumulation-mode particle number concentrations (PNCs) in Qingdao, respectively. **b, d, f** Show the corresponding results for Beijing. Percentages in each panel summarize the cumulative relative importance of meteorological variables, trace gases and background particulate matter within the corresponding mode. The nucleation mode is defined operationally as particles smaller than 25 nm; the operational nucleation-mode ranges were 5.6–25 nm in Beijing and 10.8–25 nm in Qingdao.



Mode-resolved attribution separates nucleation-mode variability from growth-mode variability

To connect the size-resolved results to the NPF formation–growth sequence, we further grouped particles into nucleation, Aitken and accumulation modes and trained separate XGBoost models for each mode in both cities (Fig. 3 and Table S4). This mode-based analysis condenses the continuous size-resolved pattern into process-relevant particle classes and enables direct comparison of environmental contributions during nucleation and subsequent growth.

In the nucleation mode, meteorological variables accounted for the largest share of model-explained variability in both cities. Their combined contribution was ~58% in Qingdao and ~56% in Beijing (Fig. 3a, b). RH was consistently the leading meteorological contributor, although its effect depended on humidity level. Low RH was associated with positive SHAP contributions to nucleation-mode PNCs, whereas higher RH was associated with smaller or negative contributions, with suppression beginning above ~50% RH in Qingdao and above ~20% RH in Beijing (Figs. S6 and S7). Solar radiation was also important in Beijing, ranking second with a relative contribution of ~16%, consistent with the role of photochemistry in

producing low-volatility vapors. These results indicate that meteorological variability provides the strongest model-attributed signal in nucleation-mode PNCs.

The attribution pattern changed in the Aitken and accumulation modes. In Qingdao, the leading contributors to Aitken-mode PNCs included NO₂, temperature, NO and PM_{2.5} (Figs. 3c and S8). In Beijing, meteorological variables explained only ~39% of Aitken-mode variability, while pollution-related variables, mainly PM_{2.5}, O₃ and NO₂, together accounted for ~61% (Figs. 3d and S9). In the accumulation mode, Qingdao showed an even clearer pollution-related signal: NO₂ alone explained ~37.8% of the modeled variability, with CO and PM_{2.5} also contributing substantially (Figs. 3e and S10). Beijing differed from Qingdao in that RH remained the largest contributor to accumulation-mode variability, accounting for 51%, while NO₂, SO₂ and O₃ also made positive contributions (Figs. 3f and S11).

These mode-resolved results refine the size-resolved pattern. In Qingdao, attribution shifted clearly from meteorology-dominated nucleation-mode variability to pollution-related accumulation-mode variability. In Beijing, meteorological variables remained important across a wider size

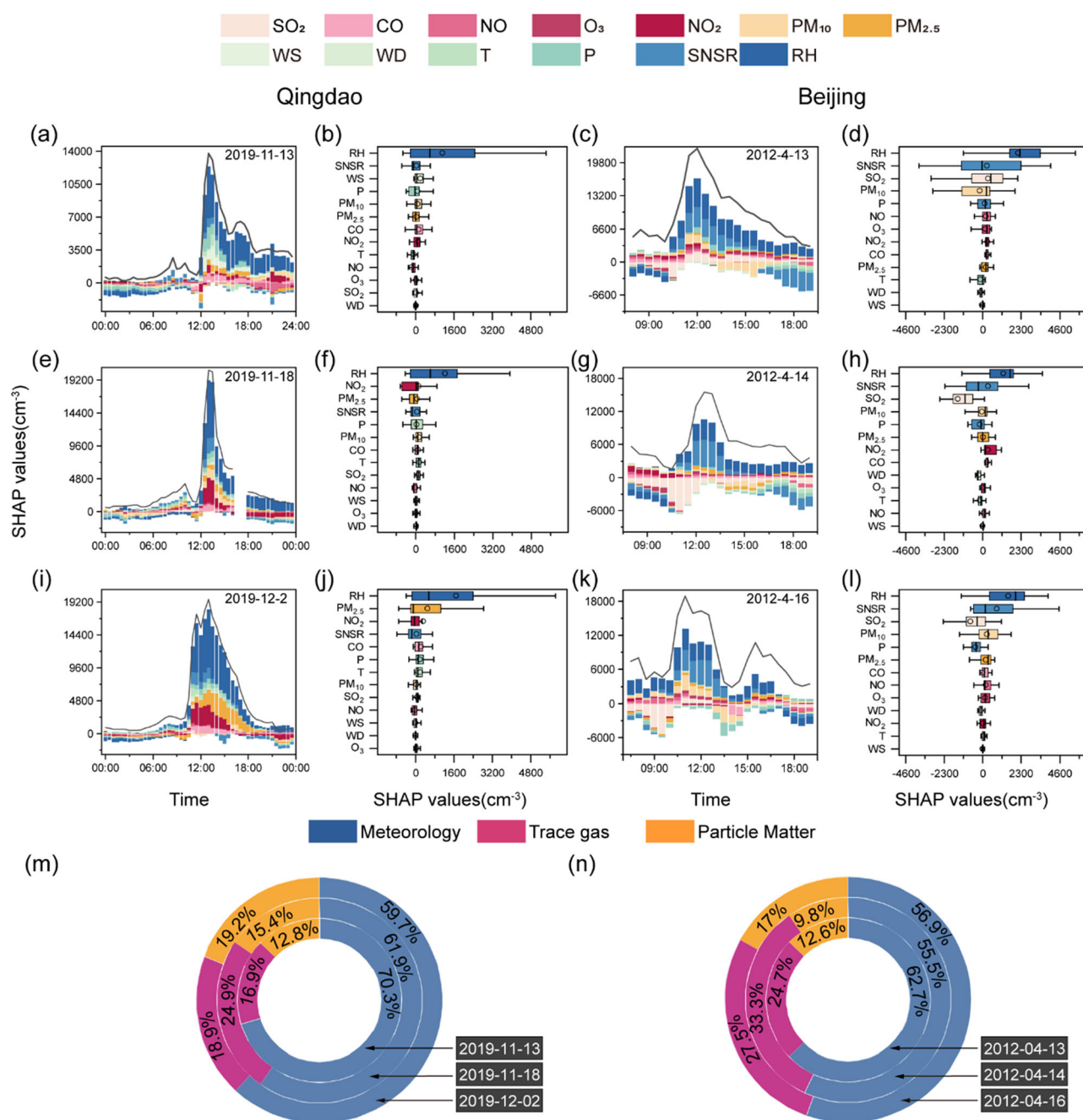


Fig. 4 | Time-resolved SHAP attribution during representative urban particle-burst events. a–l Show 30 min averaged SHAP values of the leading predictors for nucleation-mode particle number concentrations (PNCs) during representative NPF events in Qingdao and Beijing. Positive SHAP values indicate contributions that increase the modeled PNC relative to the model baseline, whereas negative values indicate decreasing contributions. For each event, the adjacent box plot summarizes the distribution of signed SHAP values for the displayed predictors,

which are ordered by their mean absolute SHAP values; boxes indicate the inter-quartile range, center lines indicate medians, circles indicate means, and whiskers extend to 1.5 times the interquartile range. **m, n** Show the cumulative relative importance of meteorological variables, trace gases and background particulate matter for nucleation-mode PNC on each event day in Qingdao and Beijing, respectively.

range, while pollution-related variables were most prominent in selected growth-related modes. Thus, the central result is not a universal switch from meteorology to pollution, but a stage-dependent redistribution of environmental contributions whose expression depends on urban background conditions.

Meteorological contributions intensify during NPF event periods

The preceding analyses describe average size- and mode-resolved attribution across the full datasets. To examine whether these patterns became more pronounced during active NPF periods, we analyzed

representative event days using 30 min averaged SHAP time series for nucleation-mode PNC (Fig. 4). Meteorological variables accounted for approximately 59–70% of nucleation-mode attribution across the representative Qingdao events (Fig. 4m and S12). RH was frequently the leading contributor. On 13 November, RH below 50% was associated with positive SHAP contributions of up to approximately +5000 cm⁻³ (Fig. 4b), whereas pressure was the leading meteorological contributor on several other Qingdao event days (Fig. S13). In Beijing, meteorological variables similarly accounted for approximately 56–63% of nucleation-mode attribution during the representative events, with RH

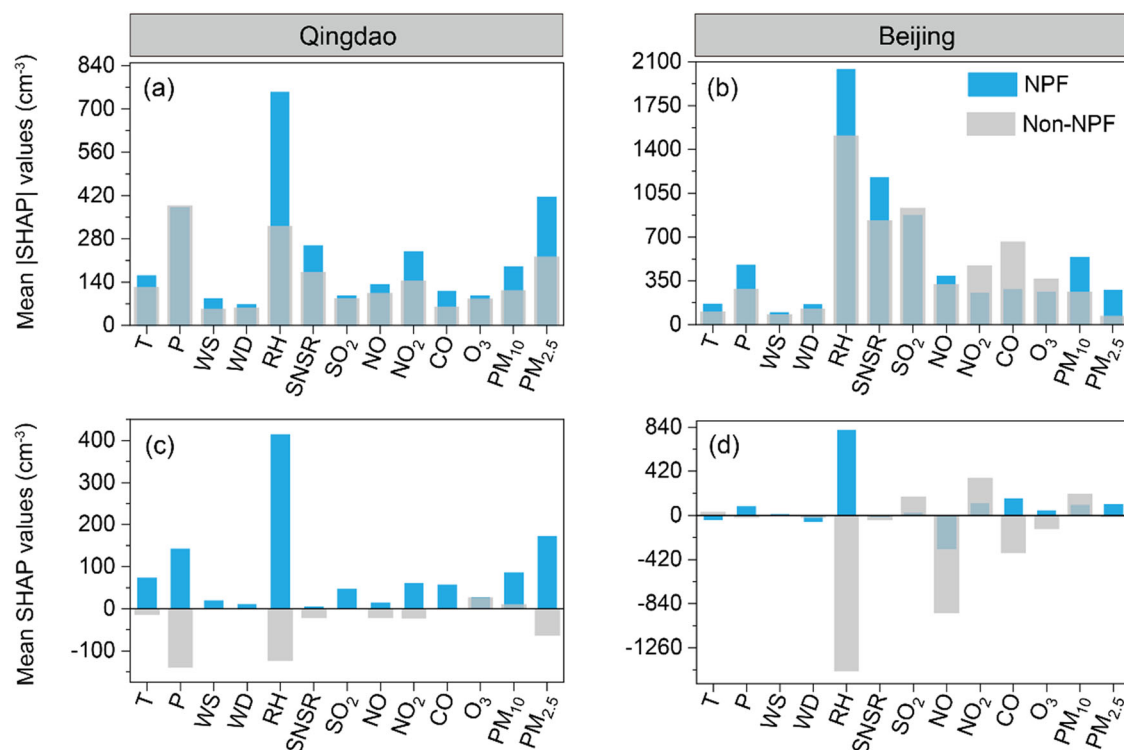


Fig. 5 | Event–non-event differences in SHAP attribution for nucleation-mode particle number concentrations. **a, b** Show mean absolute SHAP values of individual predictors for nucleation-mode particle number concentration (PNC) on event and non-event days in Qingdao and Beijing, respectively. **c, d** Show the corresponding mean SHAP values, which indicate the direction and magnitude of predictor contributions relative to the model baseline. Positive mean SHAP values

indicate net positive contributions to modeled nucleation-mode PNC, whereas negative values indicate net negative contributions. Event and non-event classifications followed the operational criteria described in “Methods,” requiring the emergence of a distinct mode within the smallest instrumentally resolved interval below 25 nm and its coherent displacement toward larger diameters over several hours.

and surface net solar radiation among the leading contributors (Figs. 4n and S14).

Pollution-related variables also contributed during individual events, but their importance was more event dependent. In Qingdao, $PM_{2.5}$ ranked among the three leading predictors on 18 November and 2 December, although particle-loading variables together accounted for less than 25% of the total attribution. Combined with the negative $PM_{2.5}$ contributions to nucleation-mode PNC shown by the SHAP dependence analysis (Fig. S6), this pattern is consistent with an influence of pre-existing particle and vapor sinks under higher aerosol-loading conditions. However, $PM_{2.5}$ may also represent a broader polluted-air-mass state that covaries with precursor availability and atmospheric mixing. In Beijing, SO_2 ranked among the leading event-day predictors, whereas it did not show comparable importance in Qingdao, consistent with the higher sulfur-related background represented by the Beijing campaign. These interpretations remain attribution based and should be regarded as process-consistent associations rather than direct evidence of individual chemical pathways.

The event–non-event comparison further supports a stronger meteorological signal during classified NPF events. In Qingdao, the mean absolute SHAP value of RH was 754 cm^{-3} on event days, 134% higher than on non-event days; the corresponding values for SNSR and $PM_{2.5}$ increased by 49% and 86%, respectively (Fig. 5a). In Beijing, the mean absolute SHAP values of RH and SNSR increased by 35% and 41%, respectively, on event days (Fig. 5b). The mean signed SHAP value of RH also shifted from negative on non-event days to positive on event days at both sites (Fig. 5c, d). This sign change should not be interpreted as a reversal of a single RH-dependent mechanism. Rather, it indicates that RH was embedded in different combinations of boundary-layer conditions, pre-existing particle sinks, hygroscopic responses, and precursor-related atmospheric states during event and non-event periods.

Because primary combustion particles can also contribute to nucleation-mode PNC in urban environments, we further stratified the SHAP values by event status and operationally defined local-time windows to examine this potential influence (Fig. S15). NO and NO_2 retained measurable model attribution during some commuting-time windows, whereas RH and SNSR generally remained among the leading predictors during midday periods on NPF event days. This result indicates that combustion-related atmospheric conditions were associated with part of the nucleation-mode variability, but the available proxies do not permit source-resolved separation of primary traffic particles from secondary particles formed in the atmosphere.

Finally, to complement the attribution-based analyses with a conventional event-level descriptor, apparent particle growth rates were calculated for all classified NPF events (Table S5). Apparent GR values ranged from 2.91 to 3.96 nm h^{-1} for the three Beijing events and from 0.73 to 3.25 nm h^{-1} for the five Qingdao events. These estimates provide a quantitative description of the observed modal displacement but do not identify the vapors or processes responsible for particle growth. Because only eight events were available and the campaigns differed in season, instrumentation, and atmospheric background, the GR values are interpreted descriptively rather than as evidence for a statistically general relationship between particle growth and any individual environmental variable. Taken together, the event-level analyses show that meteorological variables, particularly RH and solar radiation, were most strongly associated with observable nucleation-mode enhancements, whereas pollution-related variables contributed more conditionally during individual events and became more important during later modal development and at larger particle sizes.

Discussion

This study provides a size- and stage-resolved attribution of environmental controls on urban NPF. Across two contrasting urban environments,

meteorological variables, particularly relative humidity and solar radiation, accounted for the largest share of model-explained variability in nucleation-mode particles, and their contributions increased during NPF event periods. In contrast, pollution-related variables, including NO_2 , O_3 , $\text{PM}_{2.5}$ and CO , gained importance in Aitken and accumulation-mode particles, although the strength and size range of this transition differed between Qingdao and Beijing. These results show that the relative importance of meteorology and pollution is not fixed across the particle size distribution. Instead, urban NPF is best interpreted as a stage-dependent process in which meteorological conditions shape the appearance of nucleation-mode bursts, whereas pollution-related environmental states modulate subsequent growth.

These findings support and refine previous field observations. Earlier studies have shown that NPF occurrence is often favored by strong solar radiation, low relative humidity and low background aerosol loading, while growth rates and particle survival depend on condensable vapors and pre-existing particle sinks^{12–14,21}. Large multi-site analyses have also emphasized that the direction and magnitude of meteorological and compositional effects vary among locations¹⁴. Our contribution is to move beyond event-level associations by quantifying how these environmental contributions partition continuously across particle sizes and particle modes. The results do not overturn the established view that NPF requires chemical precursors; rather, they clarify when and where measured meteorological and pollution-related variables become most important in explaining observed particle number variability.

We use “gatekeeper” in an observational sense: favorable meteorological states mark the conditions under which precursor chemistry is more likely to appear as an observable nucleation-mode particle burst. Low relative humidity can reduce particle- and vapor-sink effects and is frequently associated with conditions favorable for NPF, whereas strong solar radiation enhances photochemical production of low-volatility vapors that can participate in cluster formation and early growth^{14–18}. In our analysis, these conditions were reflected by large positive SHAP contributions from RH- and radiation-related variables during nucleation-mode events. This interpretation should be understood as process-consistent attribution rather than direct causal proof. The chemical basis of nucleation remains essential, including sulfuric acid, bases and low-volatility organic compounds^{7–13}. The present results therefore suggest that meteorology helps set the conditions under which available precursor chemistry becomes expressed as an observable particle burst, rather than replacing precursor chemistry itself.

The increasing importance of pollution-related variables at larger particle sizes is consistent with current understanding, but it should not be interpreted as evidence for a single particle-growth pathway. NO_2 , O_3 , CO , and $\text{PM}_{2.5}$ are not direct measurements of the molecular species responsible for particle growth; rather, they provide indirect information on emissions, oxidative conditions, combustion influence, atmospheric aging, and pre-existing aerosol loading (Table S6). The meteorology–pollution grouping is therefore analytical rather than mechanistic, because meteorological and chemical processes remain closely coupled in the atmosphere. Accordingly, the larger SHAP contributions of pollution-related variables in the Aitken and accumulation modes indicate that model predictions at these sizes were more sensitive to pollution-related atmospheric states than were predictions of nucleation-mode PNC. This pattern is consistent with studies showing that urban emissions can supply condensable organic vapors and that combustion-related chemistry may contribute to nanoparticle growth and survival^{21–26}. Conversely, $\text{PM}_{2.5}$ showed predominantly negative contributions to nucleation-mode PNC in the SHAP dependence analysis. Because $\text{PM}_{2.5}$ is a bulk aerosol-loading proxy rather than a direct measure of condensation or coagulation sink, this result is consistent with stronger pre-existing particle and condensable-vapor sinks but does not establish either process independently. Together, these patterns point to competing size-dependent influences: polluted atmospheric states may coincide with stronger sinks that limit the observable emergence of new particles, while also providing precursor- and oxidation-rich conditions associated with particle growth at larger sizes.

Primary ultrafine particles introduce an additional interpretive constraint for nucleation-mode PNC. Because direct traffic counts, black-carbon measurements, and source-resolved particle composition were unavailable, NO and NO_2 were treated as traffic- and combustion-related proxies, whereas CO was interpreted as a longer-lived indicator of combustion influence, regional background, and dilution rather than as a traffic-specific tracer. The time-window-stratified analysis showed that NO and NO_2 retained measurable model attribution during some operationally defined commuting-time windows, while RH and surface net solar radiation generally remained among the leading predictors during midday periods on NPF event days. This analysis therefore indicates a possible combustion-related influence but does not permit source-resolved separation of primary traffic particles from secondary particles formed in the atmosphere. The apparent growth-rate analysis provides a complementary quantitative description of event-level modal displacement. Differences in apparent GR may reflect the combined effects of condensable-vapor availability, oxidation chemistry, humidity, pre-existing particle sinks, mixing, and air-mass evolution. However, the limited number of events and the differences in season, instrumentation, and atmospheric background between the two campaigns do not support attributing the observed GR differences to any individual meteorological or pollution-related variable. Overall, Table S6 and the event-level analyses provide process-consistent context for the size- and mode-resolved attribution, but they do not isolate individual molecular pathways. Direct mechanistic confirmation would require measurements of sulfuric acid, basic compounds, oxygenated organic molecules, particle chemical composition, and size-resolved condensation sink.

The contrasting attribution patterns in Qingdao and Beijing indicate that the meteorology–pollution balance was context dependent rather than uniform across particle sizes. In the Qingdao campaigns, which generally represented lower-pollution coastal conditions, the attribution shifted relatively clearly from meteorology-dominated nucleation-mode variability to stronger pollution-related contributions in the accumulation mode. Beijing showed a different size dependence: meteorological variables dominated the nucleation mode, pollution-related variables accounted for the larger share of attribution in the Aitken mode, and RH again became the leading predictor in the accumulation mode. The two datasets therefore provide useful contrasting observational cases, but the differences should not be interpreted as a controlled city-to-city response to pollution background alone.

Campaign-level statistics show that the datasets differed in modal PNCs, pollutant concentrations, RH, surface net solar radiation, season, observation year, measurement duration, and particle-sizing instrumentation (Table S7). The observed contrasts may therefore reflect the combined influence of precursor availability, pre-existing particle sinks, boundary-layer conditions, transport, and campaign-specific sampling characteristics. Changes in wind and dilution may also have altered the sampled atmospheric background, precursor-related conditions, and particle loading, although air-mass origins cannot be resolved without back-trajectory or source-resolved transport analysis (Figs. S12 and S14). Similarly, the higher SO_2 attribution in Beijing is consistent with a stronger sulfur-related atmospheric influence during that campaign, but it does not demonstrate sulfuric-acid-driven nucleation in the absence of direct H_2SO_4 or molecular-cluster measurements. Accordingly, the Beijing–Qingdao comparison is used to examine size- and stage-dependent attribution under contrasting atmospheric settings, while the seasonal statistics are used to define the observational context rather than to isolate a season-only effect. The central conclusion is therefore restricted to the measured campaign space and does not separate independent effects of location, season, instrumentation, or pollution background. This interpretation is consistent with broader evidence that NPF and particle-growth pathways vary substantially among atmospheric environments².

The results provide observational benchmarks for interpreting urban aerosol observations and evaluating size-resolved aerosol-model sensitivities. First, they show that total particle number concentration or binary event classification alone is insufficient to describe urban NPF. The same

environmental variable can differ in importance across nucleation, Aitken, and accumulation modes, and its contribution may change between event and non-event states. Second, the size-resolved SHAP framework provides a way to compare modeled particle-number variability with physically interpretable environmental variables. Such information can help assess whether size-resolved aerosol models show the expected sensitivities to meteorology and pollution-related proxies, even before all molecular pathways are simulated explicitly. Third, the results suggest that anticipating nucleation bursts requires high-quality meteorological information, whereas interpreting subsequent growth requires better representation of emissions, oxidation chemistry, and pre-existing particle sinks. These implications are best viewed as process-level constraints for observation-based interpretation and model sensitivity evaluation, not as a direct forecast system or a direct estimate of climate effects.

Several limitations define the scope of these conclusions. SHAP quantifies how measured variables contribute to the predictions of a trained model; it does not establish causal pathways on its own^{31,32}. The available covariates also do not include several key molecular precursors and growth agents, such as sulfuric acid, ammonia, amines, iodine species, highly oxygenated organic molecules or size-resolved condensation sink. As a result, the pollutant variables used here should be interpreted as proxies for broader chemical and physical environments. In addition, the Beijing and Qingdao datasets differ in observation period, instrumentation and urban background, which limits direct generalization to all cities. Direct traffic counts, black-carbon measurements, size-resolved condensation-sink estimates and air-mass back trajectories were not available. The present analysis therefore cannot provide traffic-source apportionment, isolate individual molecular growth pathways or quantitatively separate local processing from regional transport. These limitations do not weaken the central attribution result, but they define its proper interpretation: the study identifies robust, size- and stage-dependent statistical contributions within the measured environmental space.

Future work can build on this framework in three directions. First, coupling size-resolved explainable ML with direct measurements of H_2SO_4 , NH_3 , amines, VOCs, oxygenated organic molecules and condensation sink would help connect environmental attribution with molecular mechanisms. Second, applying the same framework to longer multi-city datasets would test whether the meteorology-to-pollution transition identified here is common across urban environments or specific to particular pollution and climate regimes. Third, integrating SHAP-based environmental sensitivities with process models could help assess whether aerosol microphysical models reproduce not only particle number concentrations, but also the correct size-dependent responses to meteorology and emissions. Such integration would make explainable ML a complement to mechanistic modeling rather than a replacement for it.

Overall, this work shows that the environmental controls of urban NPF are best understood as size- and stage-dependent attributions rather than a single meteorology-versus-pollution rule. Meteorology provides the strongest model-explained signal for observable nucleation-mode bursts, whereas pollution-related conditions become more important during growth, with city-specific expression. This attribution framework offers observational benchmarks for size-resolved aerosol-model sensitivities while linking routine urban observations to process-level interpretation.

Methods

Sampling sites and measurements

Measurements were conducted at an urban site in Beijing (116.31°E, 39.99°N) and at two suburban sites in Qingdao (Site A: 120.69°E, 36.37°N; Site B: 120.67°E, 36.35°N) (Fig. S16). These datasets represent two contrasting urban atmospheric regimes: a heavily polluted inland megacity and a cleaner coastal urban environment. In Beijing, PNSDs were measured with a Fast Mobility Particle Sizer (FMPS; 5.6–560 nm) at 1-s resolution. After data-quality screening and synchronization with the environmental measurements, the analysis retained observations from 16 and 21–23 December 2011 and from 13–25 April 2012, excluding 20 and 24 April. In Qingdao,

PNSDs were measured with a Scanning Mobility Particle Sizer (SMPS; 10–500 nm) at 4-min resolution. The analysis retained observations from Site A on 30 July and 1–2, 4, 6–7, and 13 August 2019, and from Site B on 10 November, 12–30 November, and 1–31 December 2019, 1 January 2020, and 6–30 June 2020. Instrument details and calibration are provided in the Supporting Information. Because the Beijing and Qingdao campaigns were asynchronous and used different particle-sizing instruments, the two datasets are treated as contrasting observational cases rather than as a controlled inter-city experiment. Campaign-specific particle, pollutant and meteorological statistics are provided in Table S7.

Meteorological variables included temperature (T), RH, pressure (P), wind speed (WS), wind direction (WD), and solar radiation (SNSR). Concurrent pollutant measurements included SO_2 , NO, NO_2 , CO, O_3 , $\text{PM}_{2.5}$ and PM_{10} . Both NO and NO_2 were retained because they reflect different aspects of urban nitrogen chemistry and combustion influence. NO and NO_2 were interpreted as traffic- and combustion-related nitrogen oxide proxies, whereas CO was treated as a longer-lived proxy for combustion influence, regional background and dilution rather than as a traffic-specific tracer. Both $\text{PM}_{2.5}$ and PM_{10} were retained because fine- and coarse-mode background aerosols can affect particle dynamics differently through condensation, coagulation and deposition.

Event classification and operational size modes

Daily time-resolved PNSD contour plots were examined over the complete observation day using the same conceptual classification criteria at both sites. A day was classified as an NPF event only when both of the following conditions were satisfied: (i) a distinct particle mode emerged within the smallest instrumentally resolved size interval below 25 nm, corresponding to 5.6–25 nm in Beijing and 10–25 nm in Qingdao; and (ii) the modal diameter subsequently exhibited a coherent and sustained shift toward larger sizes over several hours, consistent with established observational criteria^{27,28}. Short-lived particle-number concentration spikes, enhancements confined to approximately constant diameters, and small-particle increases without a continuous growth trajectory were classified as non-events. A banana-shaped trajectory in the PNSD contour plot was regarded as a characteristic pattern supporting sustained modal growth, rather than as an independent classification criterion.

Because event identification was limited by the lower detection diameter of each instrument, this procedure identifies formation-and-growth events only after the developing particle mode enters the measured size range. Nucleation episodes that remain below the lower detection limit cannot be detected, and molecular nucleation itself is therefore not directly observed. Representative NPF event and non-event days are shown in Fig. S17, and the numbers of valid observation days and classified NPF event days are summarized in Table S7. For the mode-based analyses, particles were grouped into three operational size ranges: nucleation mode, defined as 5.6–25 nm in Beijing and 10–25 nm in Qingdao; Aitken mode, defined as 25–100 nm; and accumulation mode, defined as 100–500 nm. Throughout this study, “nucleation-mode particles” therefore refers to the smallest instrumentally measured particle-size range and does not imply direct observation of the molecular nucleation step.

Machine learning framework

We modeled particle number concentration in each size bin using 13 environmental predictors (Table S1). Four algorithms were evaluated: Ridge regression, a fully connected neural network, Random Forest, and XGBoost. Data were divided into training (80%) and testing (20%) subsets. Hyperparameters for Ridge regression, Random Forest, and XGBoost were optimized by grid search with 3-fold cross-validation within the training data. The neural network included two hidden layers with 256 and 128 nodes. Model performance was evaluated using R^2 and root-mean-square error (RMSE) on the test data. Because the train–test split evaluates out-of-sample reconstruction within the observed campaigns, the reported skill should not be interpreted as prospective forecasting performance under independent future conditions.

Separate models were trained for each size channel, yielding 109 models for Qingdao and 31 models for Beijing. Additional XGBoost models were trained for the total number concentrations of the nucleation, Aitken and accumulation modes in each city. Models were implemented in Python using scikit-learn, Keras, and XGBoost libraries^{37–39}. The Random Forest models followed the algorithm of Breiman⁴⁰.

SHAP-based interpretation

Because XGBoost showed the best overall predictive performance and stability, it was selected for interpretation. SHAP was used to quantify the contribution of each predictor to each model output^{31,32}. For a predicted concentration y_i ,

$$y_i = \Phi_0 + \sum_{j=1}^M \Phi_{i,j} \quad (1)$$

where Φ_0 is the mean prediction and $\Phi_{i,j}$ is the SHAP value of predictor j for sample i . Positive SHAP values indicate that a predictor increases the predicted concentration relative to the baseline, whereas negative values indicate the opposite. Global importance I_j for predictor j was calculated as

$$I_j = \sum_{i=1}^n |\Phi_{i,j}| \quad (2)$$

and relative importance RI_j was defined as

$$RI_j = I_j / \sum_{m=1}^M I_m \quad (3)$$

where n is the number of samples and m is the number of predictors. In this study, “driver” refers to the dominant environmental correlate in the model attribution rather than a directly isolated mechanistic agent. We therefore interpret SHAP results as stage-dependent statistical dominance within the observed environmental space, not as direct causal proof of a specific physicochemical pathway. The resulting global and relative importance metrics, defined in Eqs. 1–3, were used for all subsequent attribution analyses. For correlated predictors, SHAP values distribute model attribution among covarying variables; therefore, individual feature rankings are interpreted together with grouped categories and physical context rather than as isolated causal effects.

Data availability

The data supporting the findings of this study are available within the paper and its Supporting Information. Raw data and code used for the machine-learning analysis are publicly available at Zenodo: <https://doi.org/10.5281/zenodo.18536490>.

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Author contributions

All authors contributed to the work presented in this paper. S.Y. and L.D. conceived the study. S.Y., S.Y.Z., M.R.J., Y.D.W., and Y.X.L. carried out the

machine-learning and SHAP analyses. S.Y., S.Y.Z., C.G., H.H., M.F.G., L.D., and Y.J.H. analyzed and interpreted the data. S.Y., C.G., H.H., M.F.G., and L.D. contributed to scientific discussions. S.Y. and S.Y.Z. wrote the initial draft, and all authors contributed to revision. L.D., Y.J.H., and M.F.G. supervised the project. M.F.G., C.G., H.H., and L.D. reviewed the manuscript.

Competing interests

The authors declare no competing interests.

Additional information

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